By:	AKH	Date	Jun 2026
Rev.--- By:	---	Date	---
Chk:		Date	

MacTest

Macaulay Beam Analysis - Example Worksheet

105 - (2) Span Cant - Point UDL

Beam Sect =	W14x61	W14x90		E =	29,000,000 lb / in^2	Length Units	in
A =	17.90	26.50	in^2			Force Units	lb
I =	640.00	999.00	in^4				

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E
		in	in^2	in^4	lb / in^2
DO NOT DELETE ROWS (GRAPHED)	1	1200.00	17.90	640.00	2.900E+07
	2				
	3				
	4				
	5				

Supports

	Supp. No.	X	Type	Imposed Movement	
				* dY (vert)	** dr (rot)
		in	---	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	120.0	P	0.0	0.000
	2	480.0	P	0.0	0.000
	3	900.0	P	0.0	0.000
	4				
	5				

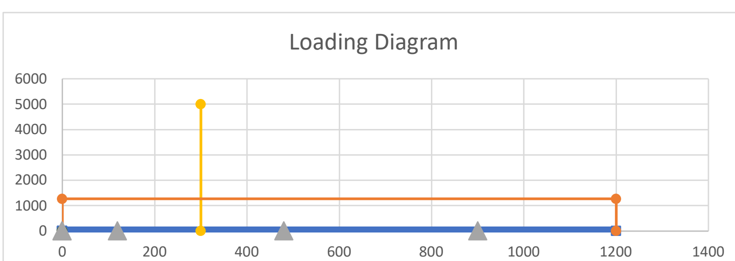
* (settlement) ** (Not Working)

Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)	Load
		in	in	lb / in	lb / in	lb / ft
DO NOT DELETE ROWS (GRAPHED)	1	0.0	1200.0	-5.1	-5.1	61.0
	2	0.0	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0	0.0
	5					

Point Loads - UNFACTORED

	Load No.	X	Py	My	My Load
		in	lb	lb-in	lb-ft
DO NOT DELETE ROWS (GRAPHED)	1	300	-5000.00	0.00	0.0
	2	0	0.00	0.00	0.0
	3				0.0
	4				0.0
	5				



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BEAM REACTIONS

Position (X)	nforced Disp (dy)	orced Rot (theta)	Vert Reac (Ry)	loment Ract (Mz)	Type
(L)	(L)	(rad)	(F)	(F-L)	(---)
120.00	0.0	0.0	3,615.5	0.0	P
480.00	0.00	0.00	4,785.5	0.0	P
900.00	0.00	0.00	2,699.0	0.0	P

Vert Reac
(kip)
3.615
4.786

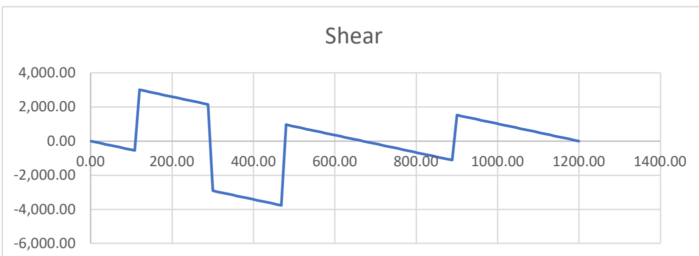
BEAM REACTIONS EQUILIBRUM CHECK

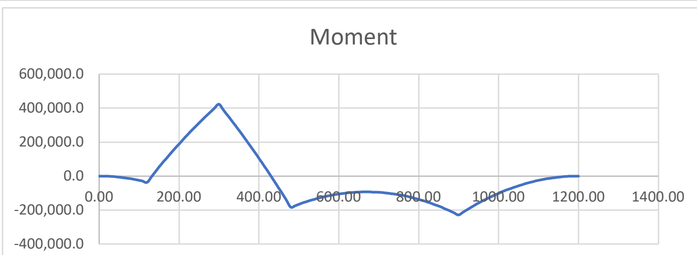
Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-11,100.0	11,100.0	0.0	OK
Sum M @ x=0	F-L	-5.1600E+06	5.1600E+06	0.0	OK

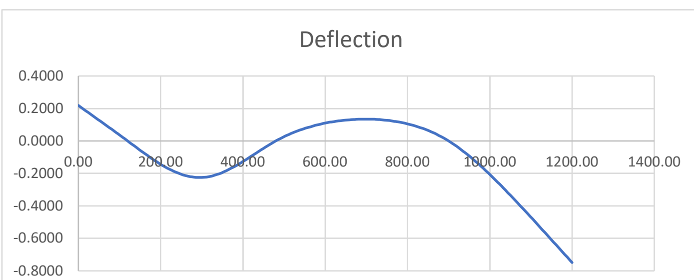
BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	120.00	3,005.5	-36,600.0	0.0000
Max -V	468.00	-3,763.5	-138,493.5	-0.0175
Max +M	300.00	-2,909.5	422,039.2	-0.2251
Max -M	900.00	1,525.0	-228,750.0	0.0000
Max Defl	1200.00	0.0	0.0	-0.7497

Max/Min Value	
(---)	
3.01	kip
-3.76	kip
35.17	kip-ft
-19.06	kip-ft
-0.750	in







INPUT SUMMARY

General Info

Beam Length:	100 ft
Section Name:	W14x61
Self Weight:	False

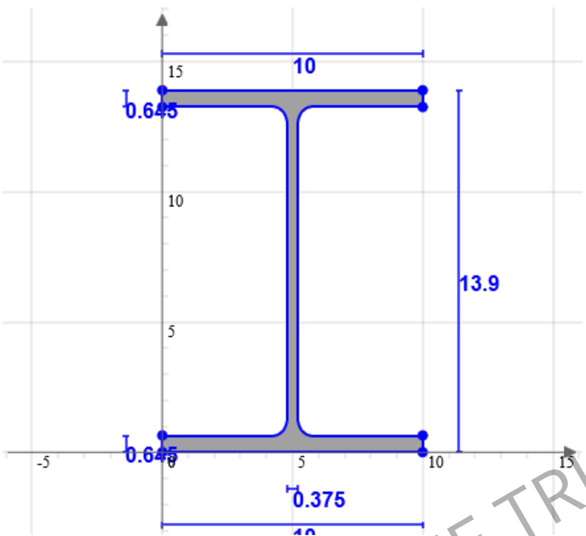
Supports

Support Type	Location
Pinned	10 ft
Roller	75 ft
Roller	40 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	25 ft	-5000 lb	DL
Distributed Load	0 ft to 100 ft	-61 lb to -61 lb	DL

Beam Section



Geometric Properties		
A	17.9	in^2
C_z	5	in
C_y	6.95	in

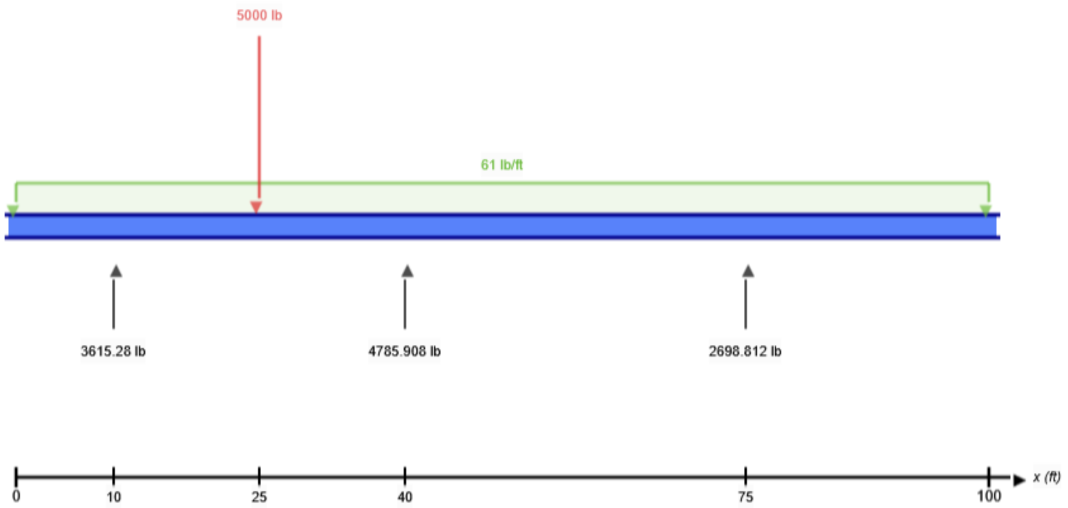
Bending Properties		
I_z	640	in^4
I_y	107	in^4

Shear Properties		
A_z	11.277	in^2
A_y	4.99	in^2

Torsion Properties		
J	2.19	in^4
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft^3)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.625	L/400
Custom Stress Limit	PASS	39 ksi	0.118	4.583 ksi
Material Yield	PASS	36 ksi	0.127	4.583 ksi
Material Strength	PASS	58 ksi	0.079	4.583 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
10	0	3615.28	0
40	0	4785.908	0
75	0	2698.812	0

Force Extremes

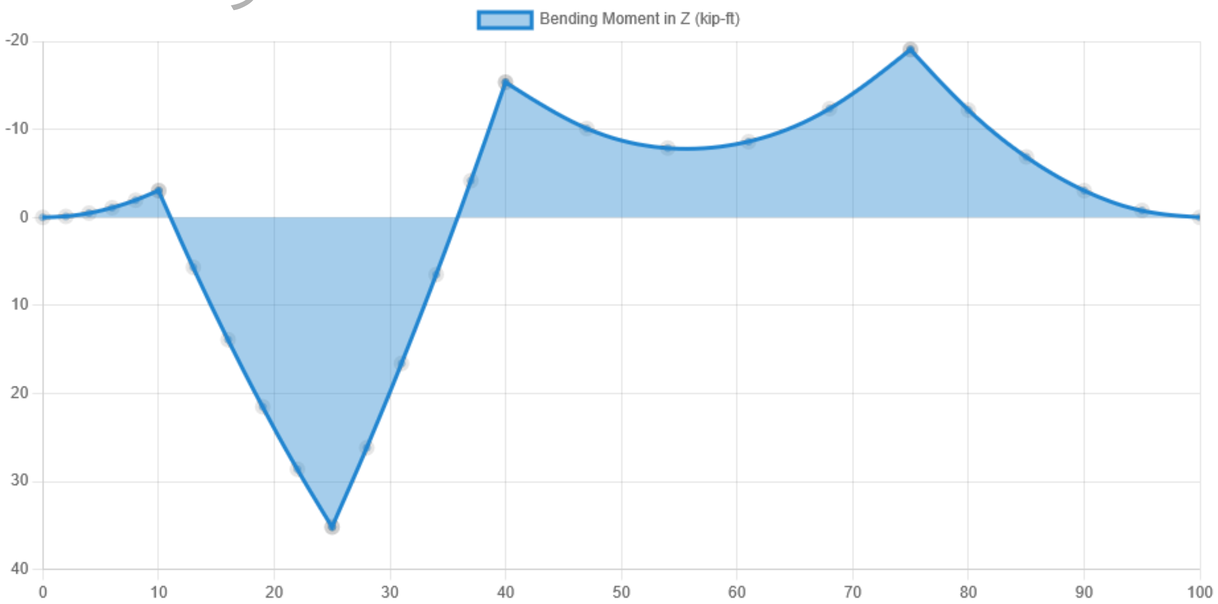
Result	Max	Min
Bending Moment	35.167 kip-ft	-19.063 kip-ft
Shear	3005.28 lb	-3824.72 lb
Displacement Y	0.219 in	-0.75 in

Stress Extremes

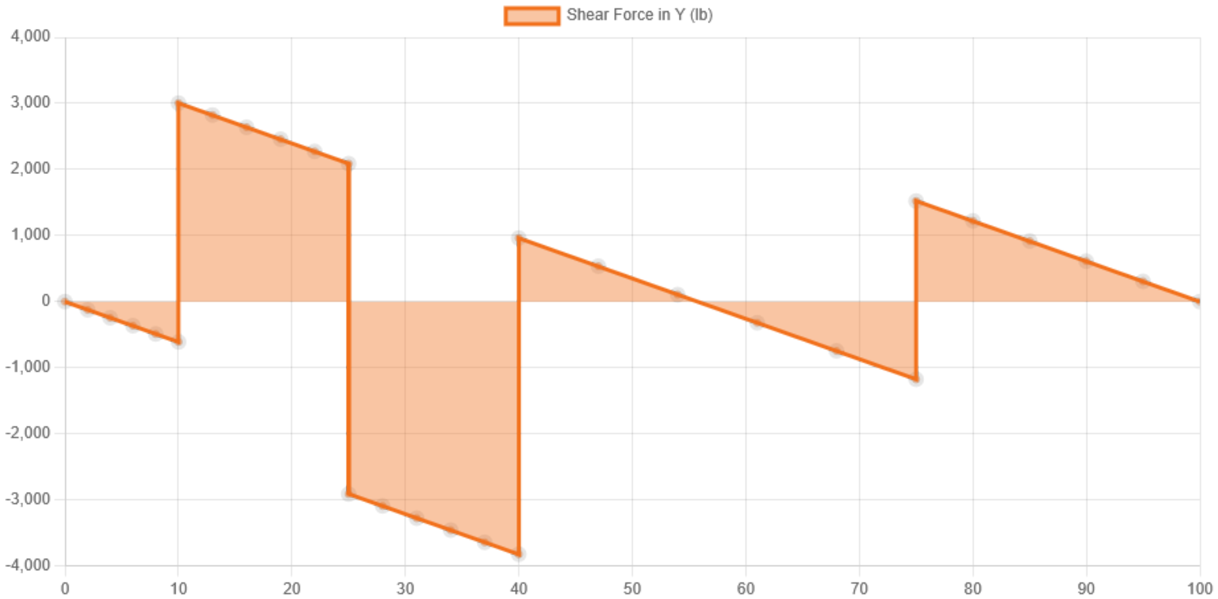
Result	Max (ksi)	Min (ksi)
Bending Stress	4.583	-4.583
Shear Stress Total	0.817	0
Max Combined Normal Stress	4.583	0
Min Combined Normal Stress	0	-4.583

DIAGRAMS

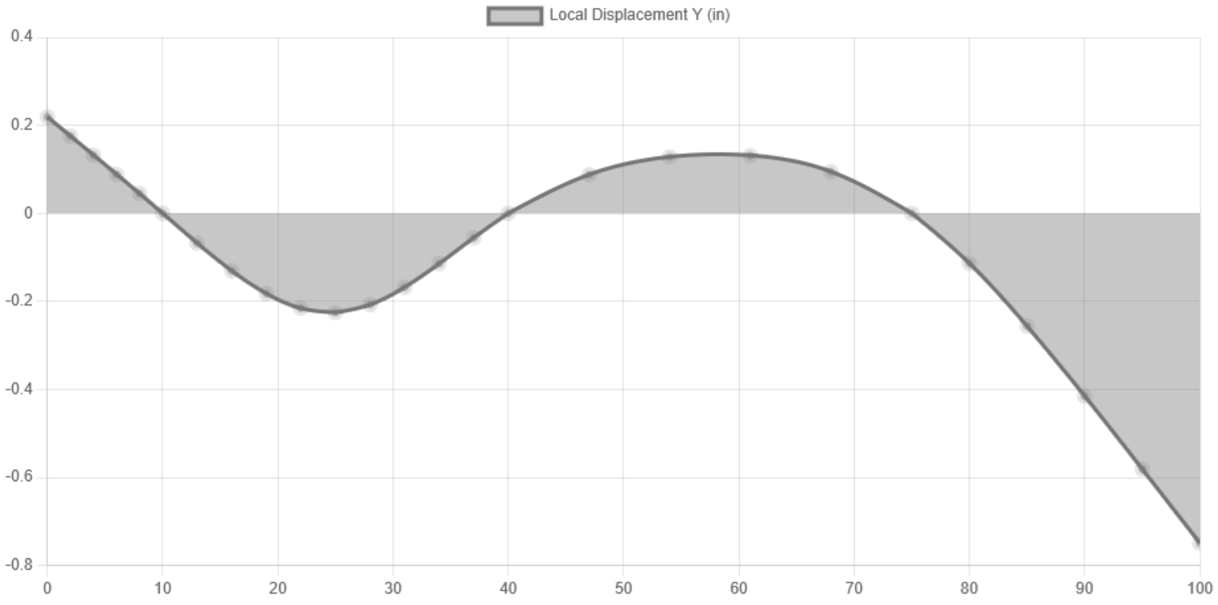
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	Total Deflection (in)	Deflection/Span
0	0.219	L/548
10	0	-
25	0.225	L/1596
40	0	-
61	0.131	L/3195
75	0	-
100	0.75	L/400

The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the **model settings.**